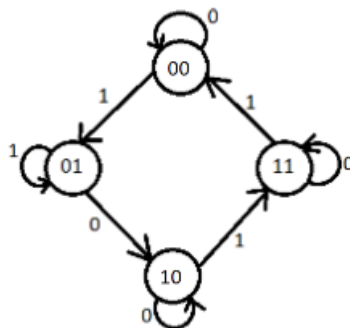


CSE260 Assignment 04
[CO3]
Total Mark: 25
Deadline: September 11, 2025; 11:59PM
Submission Link: [Click Here](#)

This assignment must be hand-written. Show ALL steps in ALL questions.

1. A toy train moves along a circular track with six stations labeled 1, 2, 3, 4, 6 and 7. The train's control system uses a **3-bit up/down counter** to keep track of the current station. Now, design the 3-bit up/down counter using SR flip flop(s).
Note: The circuit is to be designed by treating the unused states (if there is any) as Don't Care conditions.
2. Implement the following sequence using J K flip flop:
Blue → Green → Blue → Yellow → Blue → Red → Blue → Violet → Blue → Pink
3. You need to implement a **2048×16 RAM** that allows efficient data storage and retrieval.
 - a) **Draw** the diagram of a **2048×16 RAM** [using **decoder and OR gates**] showing internal details.
Note: In case of many connections, show the first, second, and last connections followed by dots in between.
 - b) For a **2048×16 RAM**, determine the following:
 - i) **Number of flip-flops** needed for the **Memory Address Register (MAR)**
 - ii) **Number of flip-flops** needed for the **Memory Buffer Register (MBR)**
 - iii) How many **address lines** are required for this RAM?
 - iv) Determine the **capacity** of this RAM in **KB**
 - v) How many **bits** can a **binary memory cell** store?
4. Given the state diagram as follows, get the sequential circuit using SR flipflop.



5. **Design** the state diagram for the following circuit diagram. Show all necessary and relevant tables and equations

